A PROJECT REPORT ON

CLASSIFICATION AND CLUSTERING OF INDIAN RAILWAY TRAINS USING MACHINE LEARNING

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Computer Applications (Honours)

Course: Machine Learning System Design & Implementation (DSC-C-BCA-473P)

Academic Year: 2025-26





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1. COLLEGE CERTIFICATE

LOKMANYA COLLEGE OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that the project report entitled **"CLASSIFICATION AND CLUSTERING OF INDIAN RAILWAY TRAINS USING MACHINE LEARNING"** is a bonafide record of the work carried out by **Rajput Prince Singh (Roll No. 83)** and submitted to **Gujarat University** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Computer Applications (Honours)** for the course **Machine Learning System Design & Implementation (DSC-C-BCA-473P)** during the academic year **2025-26**.

To the best of our knowledge, this work is original, was carried out at **Lokmanya College of Computer Applications, Ahmedabad, Gujarat**, under the guidance of **Prof. Sejal Shah**, and has not been submitted in part or in full to any other university or institution for the award of any degree or diploma.

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Date: ___ / ___ / 2026

2. ACKNOWLEDGEMENT

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First, I thank the management of **Lokmanya College of Computer Applications, Ahmedabad, Gujarat**, for providing excellent laboratory facilities, library resources, and a supportive learning environment throughout my BCA journey.

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I am deeply thankful to my project guide, **Prof. Sejal Shah, Assistant Professor**, for her valuable guidance at every stage of this work. Her suggestions helped me select the right approach for data exploration, model building, and result interpretation.

I am also grateful to **Gujarat University** for including the course **Machine Learning System Design & Implementation (DSC-C-BCA-473P)** in the BCA (Honours) curriculum under NEP 2020. This gave me the opportunity to work on a real-world classification and clustering problem using the Indian Railways dataset.

Finally, I thank my family and friends for their unwavering support, patience, and motivation throughout this project.

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3. PROJECT DEFINITION

3.1 Introduction

Indian Railways is the fourth-largest railway network in the world, spanning over 68,000 route kilometres, operating more than 13,000 passenger trains daily across 19+ administrative zones and 7,300+ stations. Every day, this network generates vast operational data — train schedules, route distances, coach configurations, journey durations, and zone-wise classifications.

Manual categorisation of trains into service tiers — **Passenger (Pass)**, **Express (Exp)**, and **Superfast (SF)** — is labour-intensive, inconsistent across zones, and difficult to scale. This project demonstrates that machine learning can automate this classification with high accuracy (91.50%) while also discovering latent operational groupings that are not explicitly labelled.

3.2 Problem Statement

The Indian Railways timetable data contains overlapping features across service tiers (e.g., Express and Superfast trains share similar distances and coach compositions), moderate class imbalance (Passenger dominates at 55.06%), and a hidden data-quality issue in the duration_m column. The challenge is to:

- **1. Multi-Class Classification:** Build an accurate multi-class classifier that predicts a train's service tier (Pass, Exp, SF) from 26 operational metrics.
- **2. Unsupervised Clustering:** Apply unsupervised clustering (K-Means and DBSCAN) to discover natural groupings of trains and isolate anomalous schedules.
- **3. Data Quality Remediation:** Diagnose and fix data-quality issues in the journey duration representation that degrade model performance.

3.3 Project Objectives

- **1. Data Engineering & EDA:** Convert the raw GeoJSON dataset (trains.json, 14.08 MB) into a structured CSV (trains.csv, 636 KB); audit constant columns; and visualise feature distributions and correlations.
- **2. Data Quality Remediation:** Diagnose and repair the broken duration_m feature via compound duration engineering total_duration = (duration_h * 60) + duration_m.
- **3. Leakage-Free Preprocessing:** Enforce strict 80/20 stratified train-test isolation (3,572 train / 894 test) before fitting StandardScaler and one-hot encoding 19 railway zones.

- **4. Multi-Class Classification:** Train and compare Dummy Baseline, KNN (k=5), Decision Tree (default and depth=10), Gaussian Naive Bayes, and SVM (RBF) on the 3 train classes.
- **5. Robust Model Validation:** Evaluate with Macro/Weighted/Micro F1, confusion matrices, One-vs-Rest ROC-AUC, 5-fold stratified cross-validation, feature importance, and error analysis.
- **6. Unsupervised Fleet Discovery:** Segment trains using K-Means (k=5, Elbow + PCA) and density-based DBSCAN (eps=7.0, min_samples=20); export labelled clusters to data/trains_with_clusters.csv.

3.4 Scope of the Project

- **In Scope:** Three-class classification (Passenger, Express, Superfast) using 26 engineered features; two unsupervised clustering algorithms (K-Means, DBSCAN); data-quality diagnostics and remediation; and reproducible pipeline with saved models and CSV outputs.
- **Out of Scope:** Live NTES delay data integration, real-time REST API deployment, and deep learning approaches (documented under Future Enhancements).

3.5 Why Machine Learning?

Machine learning is well-suited to this problem because the classification boundaries between service tiers are non-linear and threshold-based (matching real-world railway rules), unsupervised clustering can discover hidden archetypes that manual zoning cannot, and the trained pipeline can classify new train records automatically.

4. PROJECT EXPLORATION (EDA)

4.1 Dataset Overview

Table 4.1: Dataset Overview and Specifications

Field	Specification
Name	Indian Railways Dataset
Source	Kaggle — sripaadsrinivasan/indian-railways-dataset
Original File	trains.json (GeoJSON, 14.08 MB)
Converted File	trains.csv (636 KB)
Raw Records	5,208
Filtered Records	4,466 (3 classes only: Pass, Exp, SF)
Features (after one-hot)	26

4.2 Target Variable — Class Distribution

Table 4.2: Target Variable Class Distribution (N = 4,466)

Class Code	Category	Count	Share (%)	Operational Profile
Pass	Passenger	2,459	55.06%	Short-distance, frequent halts, unreserved / basic sleeper
Exp	Express	1,288	28.84%	Medium-to-long distance, mixed Sleeper & 3AC/2AC coaches
SF	Superfast	719	16.10%	High priority, fewer halts, high average speed & full AC

Observation: The dataset exhibits moderate class imbalance — the Passenger class is 3.4 times larger than Superfast. This makes raw accuracy misleading; therefore, **Macro F1 is adopted as the primary evaluation metric**.

4.3 Zone Distribution

Zone-wise train counts reveal that NR (Northern Railway), WR (Western Railway), SR (Southern Railway), ER (Eastern Railway), and CR (Central Railway) handle the highest volumes, reflecting dense commuter and intercity corridors.

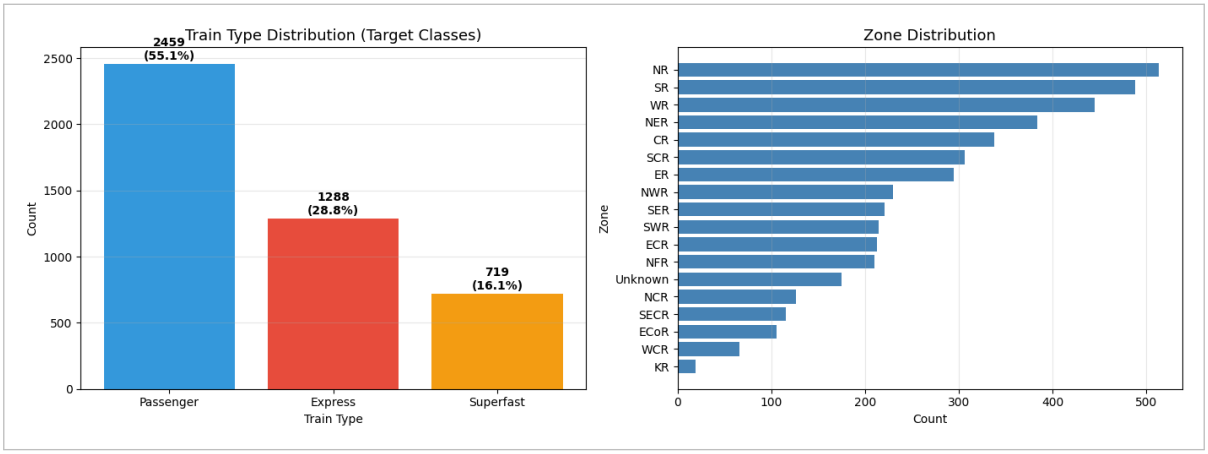


Figure 4.1: Target Class Distribution (Left) and Railway Zone Distribution (Right)

4.4 Univariate & Quartile Analysis

- **Distance Distribution:** Right-skewed distribution with a mean distance of 558 km, a median distance of 244 km, and a long upper tail extending beyond 4,000 km for long-haul Express and Superfast routes.
- **Duration Distribution (Before Fix):** Bimodal distribution bounded between 0 and 58 minutes (mean: 28 minutes, median: 30 minutes), which directly exposed the duration_m data-quality issue.

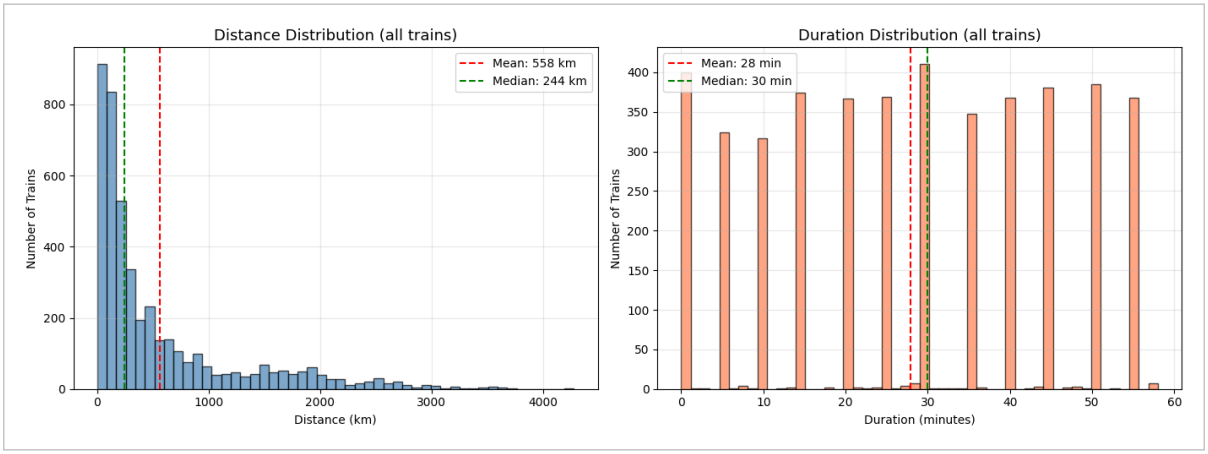


Figure 4.2: Univariate Feature Histograms Across Operational Attributes

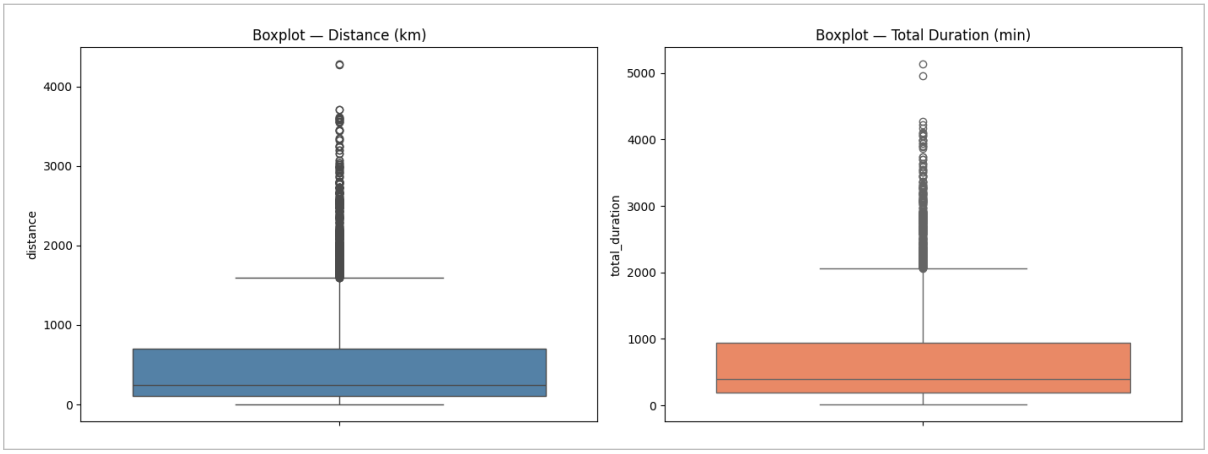


Figure 4.3: Multi-Class Boxplots Showing Quartile Spread and Route Outliers

4.5 Correlation Analysis

Table 4.3: Pearson Correlation Coefficients (r) Across Key Features

Feature Pair	Correlation (r)	Interpretation
Distance <-> Total Duration	0.98	Strong linear relationship between route length and travel time
Third AC <-> Sleeper	0.86	Long-distance overnight trains carry both Sleeper and 3AC coaches
Second AC <-> Sleeper	0.79	Upper-tier AC coaches accompany Sleeper on Express/SF routes
Distance <-> Sleeper	0.70	Sleeper coaches are attached primarily on medium/long routes
Distance <-> Third AC	0.69	Strong positive correlation between distance and 3AC availability
First AC <-> Second AC	0.39	Selective attachment of 1AC cabins on premier corridors

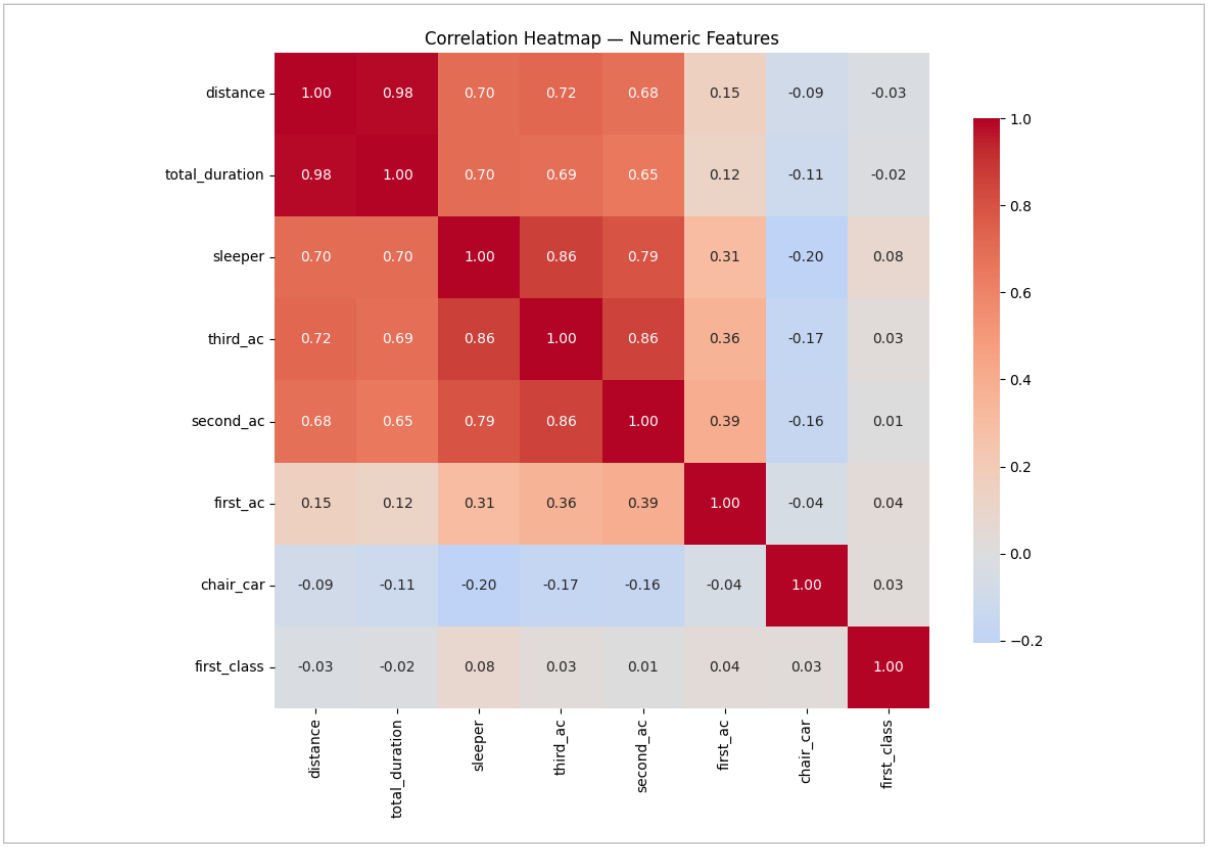


Figure 4.4: Pearson Correlation Heatmap of Numerical Features

4.6 Data Quality Issue — The Killer Finding

The **duration_m** column appeared to store total minutes, but inspection showed that it only stored the **minute remainder (0-58)** rather than total journey duration in minutes. For example, a 24-hour journey with 0 extra minutes appeared as duration_h = 24, duration_m = 0, and a 14h 30m journey appeared as duration_h = 14, duration_m = 30.

To restore the true continuous journey time across all 4,466 trains, the compound feature **total_duration** was engineered and the broken component columns were dropped:

$$total_duration = (duration_h * 60) + duration_m$$

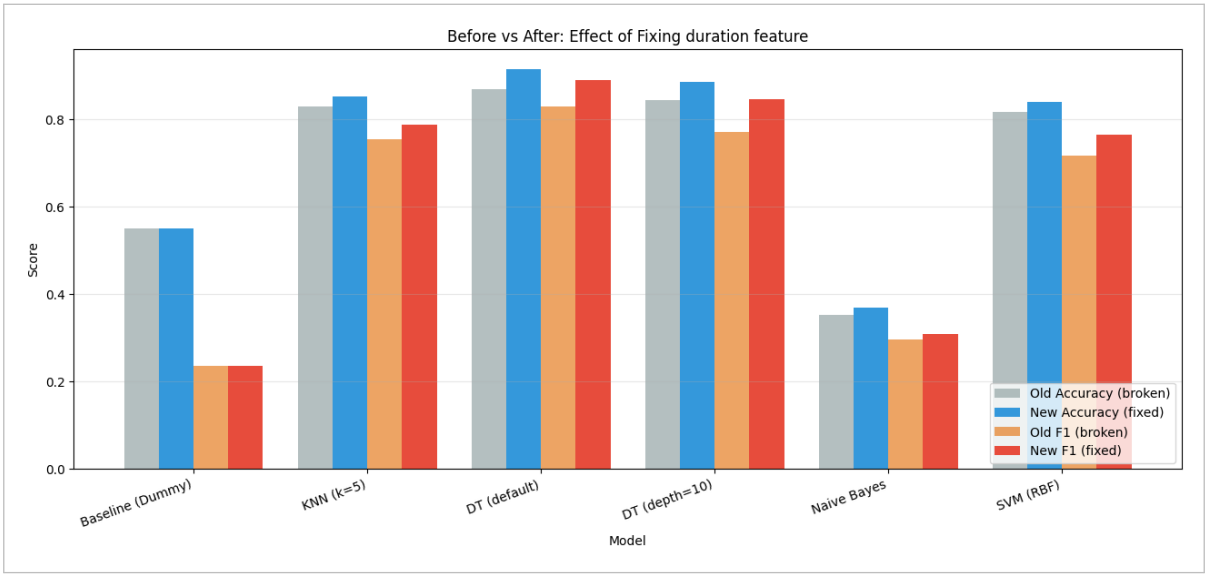


Figure 4.5: Before vs After Duration Fix — Macro F1 Improvement Across All Models

4.7 Constant Column Audit

Before one-hot encoding, all binary coach indicators (1AC, 2AC, 3AC, Chair Car, Sleeper, First Class) were audited for variance (nunique() > 1) to prevent encoding zero-variance columns.

5. EVALUATION MATRIX

5.1 Train-Test Split & Leakage Prevention

Table 5.1: Stratified 80/20 Train-Test Split

Split Partition	Size	Ratio	Role in Pipeline
Training Set	3,572	80%	Stratified model training, 5-fold CV & clustering
Testing Set	894	20%	Unseen hold-out evaluation, confusion matrix & ROC-AUC

Method: train_test_split(test_size=0.2, stratify=y, random_state=42) preserves exact class proportions in both splits. StandardScaler was fitted strictly on X_train after splitting to prevent data leakage.

5.2 Evaluation Metrics

- **1. Accuracy:** Fraction of total test trains classified correctly: (TP + TN) / (TP + TN + FP + FN). Limitation: Biased toward the majority Passenger class (55.06%).
- **2. Precision:** Of the trains predicted as class X, how many were actually class X: TP / (TP + FP).
- **3. Recall:** Of the actual class X trains, how many were correctly identified: TP / (TP + FN).
- **4. F1-Score:** Harmonic mean of precision and recall: 2 * (Precision * Recall) / (Precision + Recall).
- **5. Macro-Averaged F1 (Primary Metric):** Unweighted mean of per-class F1 scores: (F1_Pass + F1_Exp + F1_SF) / 3. Treats all classes equally — essential for imbalanced datasets.
- **6. Weighted-Averaged F1:** F1 score weighted by class support, reflecting natural fleet proportions.
- **7. Micro-Averaged Precision / Recall:** Global aggregation of True Positives across all classes (0.9150 for Decision Tree).
- **8. ROC-AUC (One-vs-Rest) & Confusion Matrix:** Measures ranking separability across probability thresholds and tabulates actual vs predicted counts in a 3x3 grid.

5.3 Impact of Data Quality Fix on Evaluation

Table 5.2: Impact of Compound Duration Fix Across All Models

Model	F1 Macro (Before)	F1 Macro (After)	Delta Gain
KNN (k=5)	0.7552	0.7886	+0.0334
Decision Tree (default)	0.8309	0.8896	+0.0587
DT (depth=10)	0.7720	0.8459	+0.0739
Naive Bayes	0.2969	0.3098	+0.0129
SVM (RBF)	0.7168	0.7651	+0.0483
K-Means Silhouette	0.1755	0.2024	+0.0269

Key Finding: A single data-quality fix improved all 5 supervised models and the clustering silhouette score — empirical proof that data quality matters more than algorithm choice.

6. MODEL BUILDING

6.1 Algorithm Selection

Table 6.1: Supervised Classification Algorithms Selected

Algorithm	Type	Reason for Selection
Dummy Classifier	Baseline	Establishes minimum benchmark floor
KNN (k=5)	Instance-based	Distance-based voting; requires feature scaling
Decision Tree	Rule-based	Interpretable threshold splits; no scaling required
Gaussian Naive Bayes	Probabilistic	Fast baseline; assumes conditional independence
SVM (RBF)	Margin-based	Non-linear kernel hyperplane; requires feature scaling

6.2 Dummy Baseline (Benchmark)

DummyClassifier(strategy='most_frequent') predicts Passenger for every test sample, achieving Accuracy = 0.5503, F1 Macro = 0.2367, and F1 Weighted = 0.3907. A model must beat 23.67% Macro F1 to be meaningful.

6.3 K-Nearest Neighbors (k = 5)

For each test point, KNN finds the 5 nearest training points by Euclidean distance and takes a majority vote. StandardScaler is required because Euclidean distance is otherwise dominated by large-magnitude features (distance in km). Result: Accuracy = 0.8523, F1 Macro = 0.7886, F1 Weighted = 0.8484.

6.4 Decision Tree Classifier (Champion Model)

Decision Tree recursively splits the feature space on axis-aligned thresholds to maximise information gain (Gini impurity). It wins because Indian Railways classifies trains via threshold rules (e.g., 3AC availability, distance, and journey duration). Default tree achieves **0.9150 Accuracy, 0.8896 F1 Macro, and 0.9146 F1 Weighted**, outperforming the pruned depth=10 tree (0.8859 Accuracy, 0.8459 F1 Macro).

6.5 Gaussian Naive Bayes

Gaussian Naive Bayes assumes features are conditionally independent given the class and follow Gaussian distributions. Because Distance, Duration, and AC coaches are strongly correlated ($r > 0.85$), Naive Bayes fails with Accuracy = 0.3702, F1 Macro = 0.3098, and F1 Weighted = 0.3636.

6.6 Support Vector Machine (RBF Kernel)

SVM finds the maximum-margin hyperplane separating classes using an RBF kernel on scaled features. Result: Accuracy = 0.8412, F1 Macro = 0.7651, F1 Weighted = 0.8343.

6.7 Model Comparison Summary

Table 6.2: Supervised Model Comparison Summary

Model	Accuracy	F1 Macro	F1 Weighted	Verdict
Dummy Baseline	0.5503	0.2367	0.3907	Benchmark
KNN (k=5)	0.8523	0.7886	0.8484	Strong
Decision Tree (default)	**0.9150**	**0.8896**	**0.9146**	**WINNER**
DT (depth=10)	0.8859	0.8459	0.8836	Runner-up
Naive Bayes	0.3702	0.3098	0.3636	Poor fit
SVM (RBF)	0.8412	0.7651	0.8343	Strong

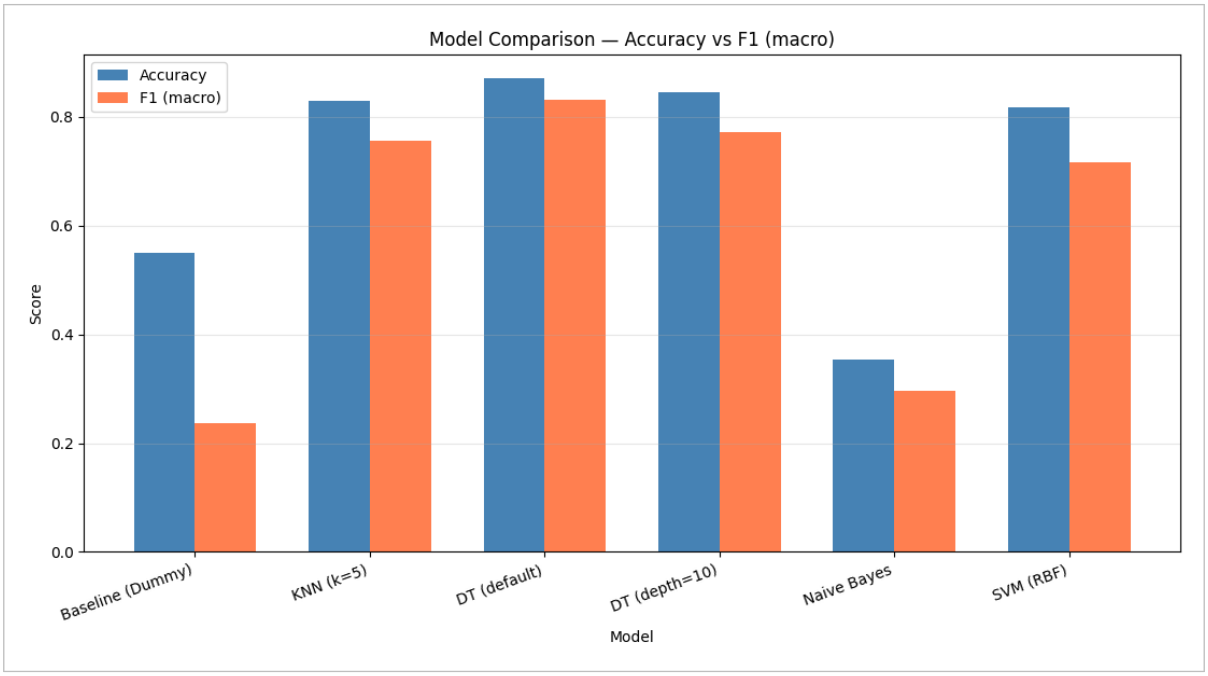


Figure 6.1: Supervised Model Comparison Bar Chart

6.8 Confusion Matrix — Decision Tree (Best Model)

Table 6.3: Confusion Matrix and Per-Class Error Rates (Decision Tree)

Actual \ Predicted	Pass	Exp	SF	Per-Class Error Rate
Pass	464	26	2	3.0% error (97.0% recall)
Exp	20	202	36	14.0% error
SF	0	32	112	17.4% error

Overall error rate is **8.50%**. Dominant confusion occurs along the SF <-> Exp boundary (25 SF -> Exp, 16 Exp -> SF), while zero Superfast trains are misclassified as Passenger.

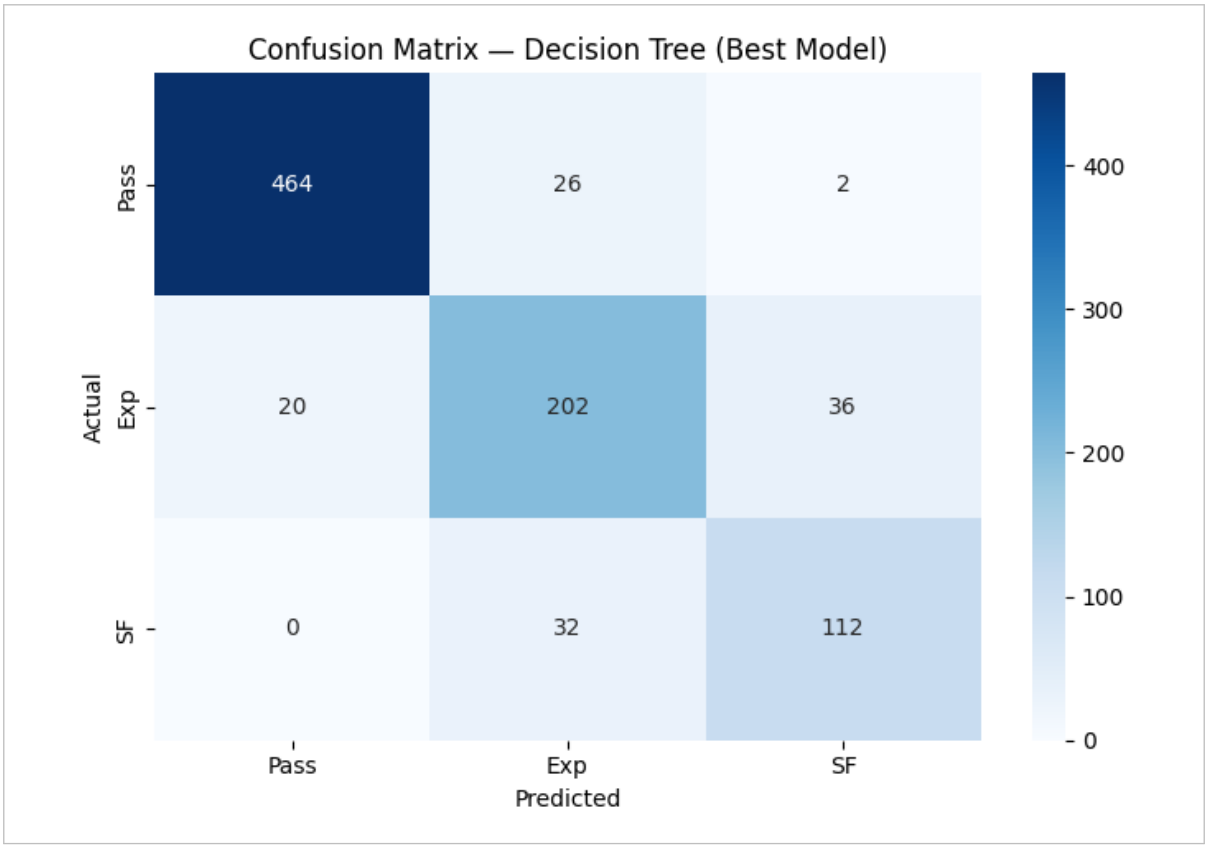


Figure 6.2: Confusion Matrix for Champion Decision Tree Model

6.9 ROC-AUC Curves

Table 6.4: One-vs-Rest Mean ROC-AUC Scores

Model	Mean AUC
KNN (k=5)	0.9355 (best)
SVM (RBF)	0.9285
Decision Tree	0.9204

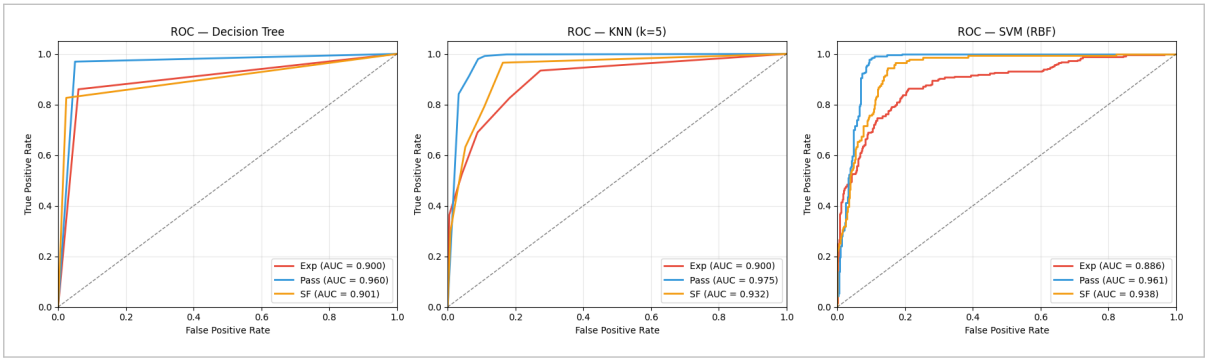


Figure 6.3: One-vs-Rest Multi-Class ROC Curves

6.10 Feature Importance (Decision Tree)

Table 6.5: Decision Tree Feature Importance Ranking

Rank	Feature	Importance	Interpretation
1	third_ac	0.3551	Primary separator between Pass and Exp/SF
2	distance	0.2226	Commuter vs long-haul separator
3	total_duration	0.1799	Engineered feature #3 — validates duration fix
4	chair_car	0.1181	Intercity day express indicator
5	first_ac	0.0133	Premium AC tier marker

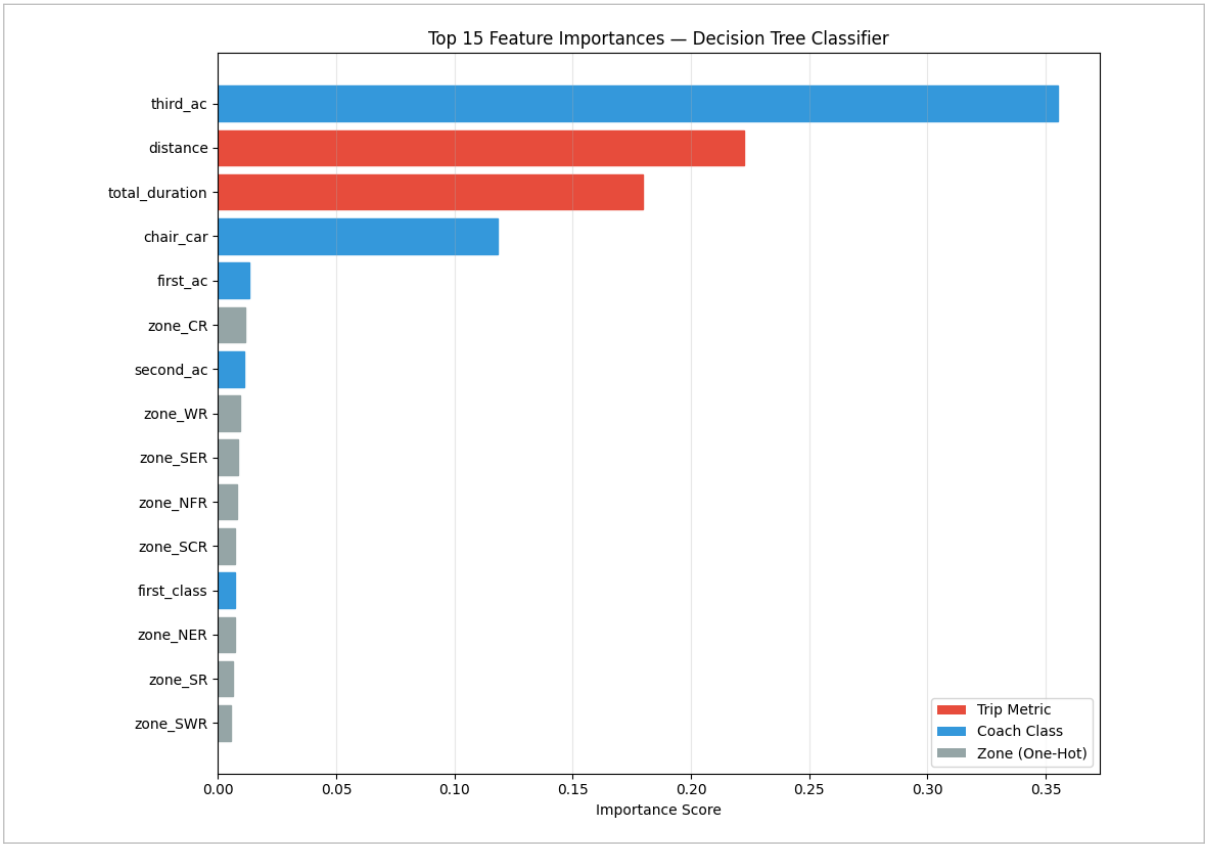


Figure 6.4: Decision Tree Feature Importance Ranking

6.11 Cross-Validation (5-Fold Stratified) & Error Analysis

Decision Tree 5-fold stratified cross-validation achieved **F1 Macro = 0.8743 +/- 0.0155**. Low variance across folds confirms stability and absence of overfitting. Error analysis confirms that misclassified trains are concentrated in the mid-distance, mid-duration region where Express and Superfast overlap operationally.

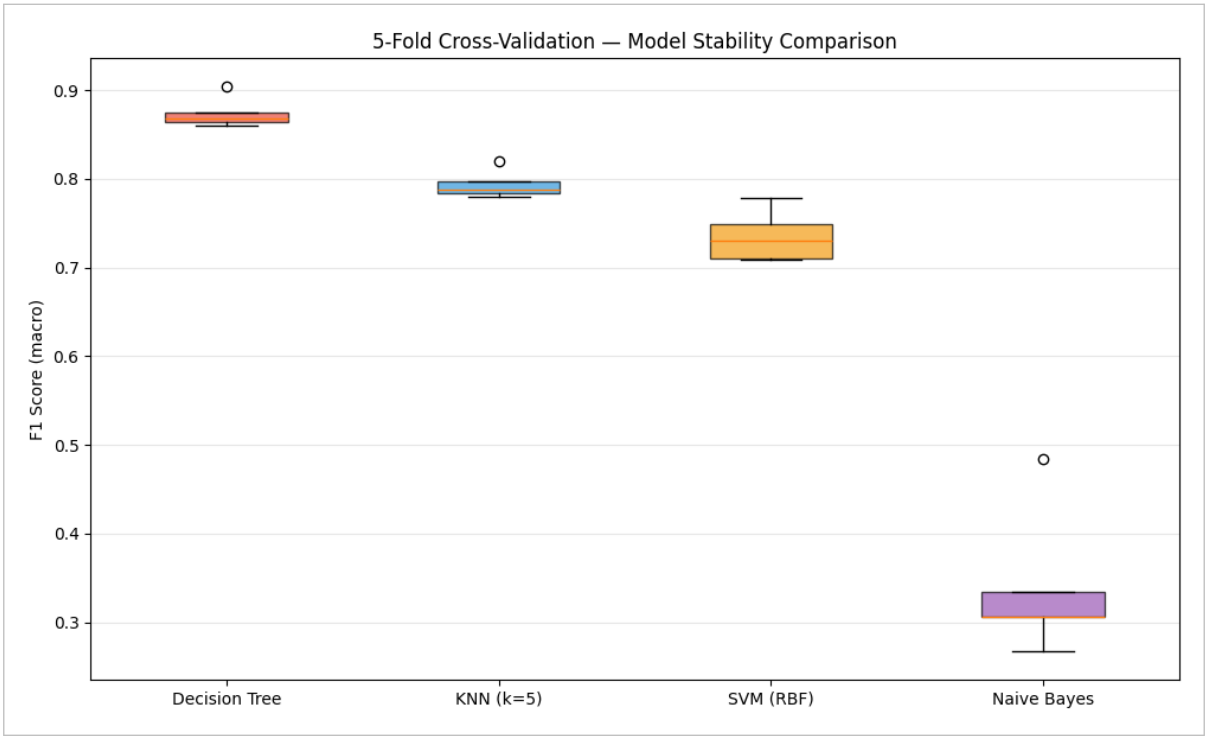


Figure 6.5: 5-Fold Stratified Cross-Validation Boxplot

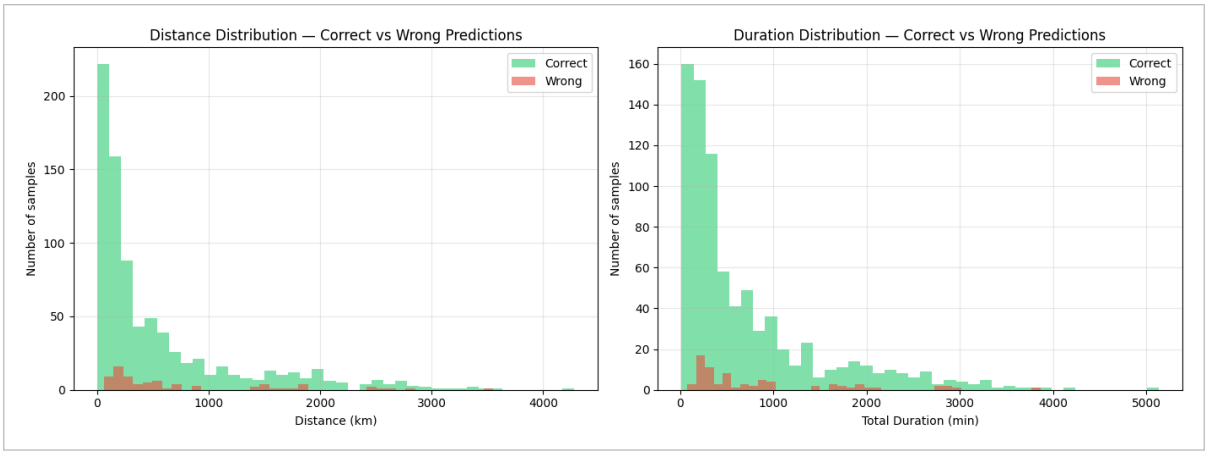


Figure 6.6: Error Analysis of Misclassified Trains

7. ADVANCED TECHNIQUES - PATTERN DISCOVERY

7.1 K-Means Clustering (k = 5)

K-Means partitioned the 3,572 scaled training records into k = 5 clusters (selected via the Elbow Method across k = 2 to 10). At k = 5, Inertia is **69,115.75** and Silhouette Score is **0.2024** (up from 0.1755 before the duration fix). For visualisation, the 26-dimensional feature space was projected to 2D using Principal Component Analysis (PCA).

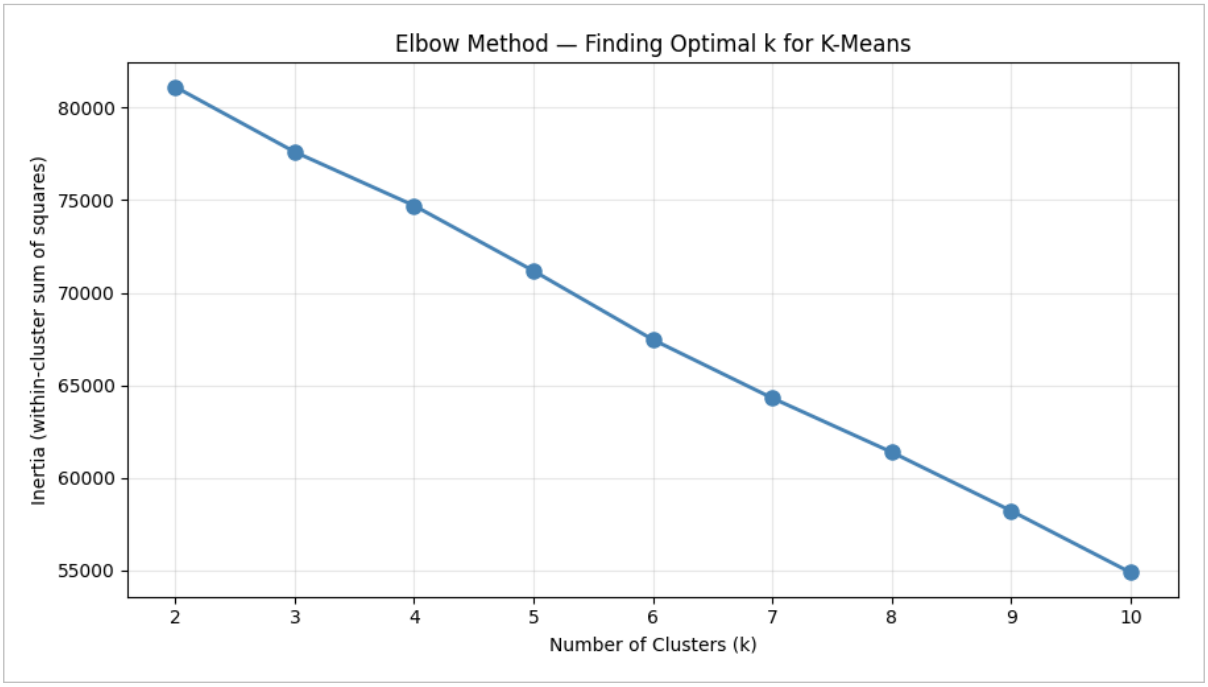


Figure 7.1: Elbow Curve for Optimal k Selection

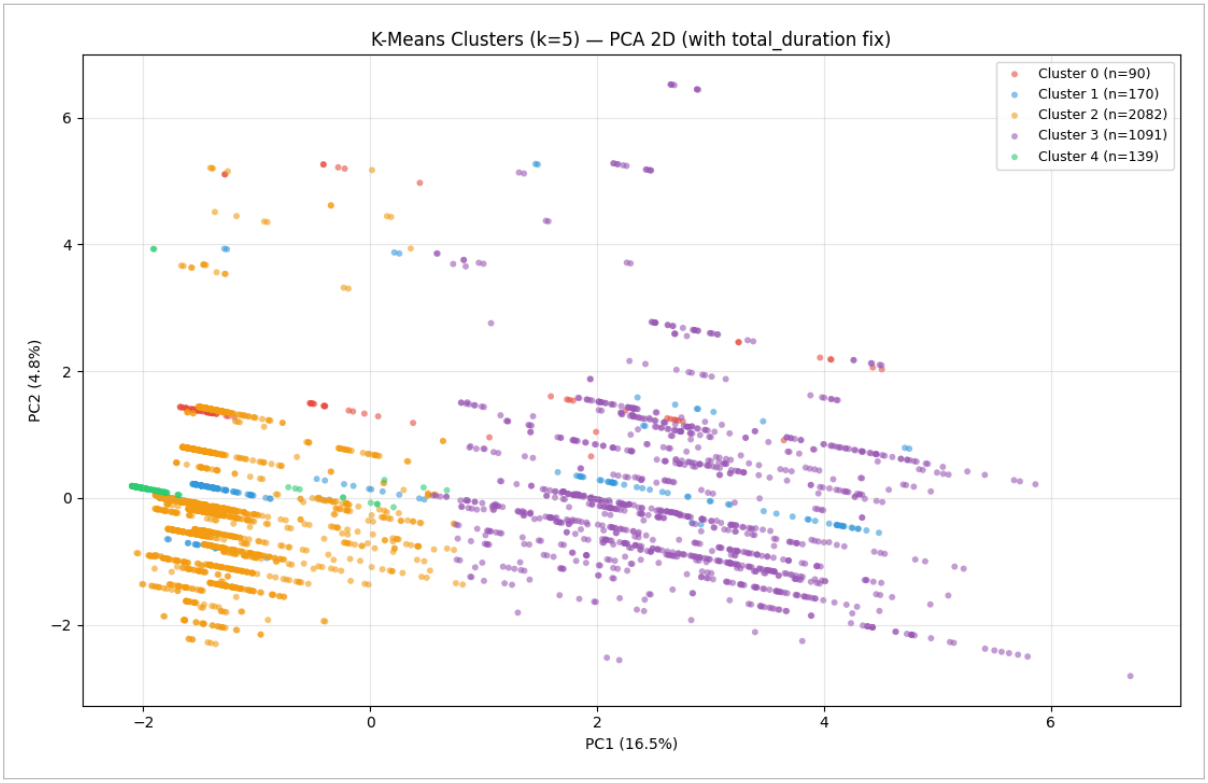


Figure 7.2: K-Means Clusters (k = 5) in 2D PCA Space

7.2 DBSCAN Clustering (eps = 7.0, min_samples = 20)

Density-based DBSCAN (eps = 7.0, min_samples = 20) discovered **4 natural clusters** with **0.4% noise (13 outlier points)** and a Silhouette Score of **0.2543** (outperforming K-Means).

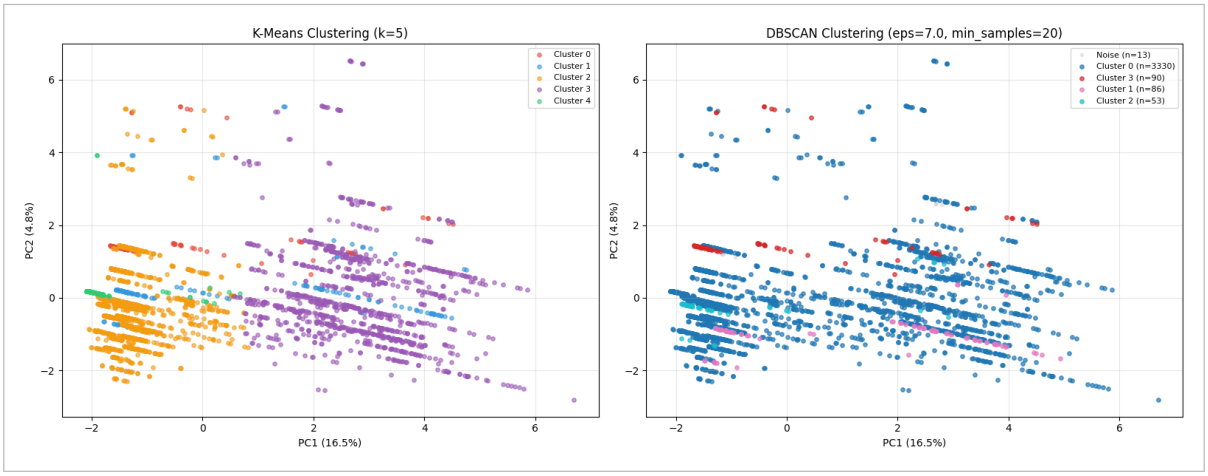


Figure 7.3: DBSCAN vs K-Means Clustering Comparison

7.3 K-Means vs DBSCAN Comparison & Cluster Interpretation

Table 7.1: K-Means vs DBSCAN Comparison

Criterion	K-Means (k=5)	DBSCAN (eps=7, min=20)
Paradigm	Centroid-based	Density-based
Clusters	5 (user-specified)	4 (natural)
Noise handling	None	0.4% (13 points)
Silhouette	0.2024	0.2543
Geometry	Spherical only	Arbitrary
Operational strength	Granular zone/tier	Anomaly isolation

Table 7.2: Interpretation of Clusters (K-Means, k = 5)

Cluster	Name	Operational Interpretation
C-0	ER-Regional Network	Eastern Railway regional fleet — medium-short distance, mixed coaches
C-1	NFR-Mixed Operations	Northeast Frontier strategic routes — hilly gradients
C-2	Short-Distance Commuter Pass	High-frequency local passenger — short route, zero AC coaches
C-3	Long-Distance AC Premium Fleet	Flagship Express & Superfast — longest distance, near-100% AC
C-4	Secondary-Zone Short Passenger	Regional branch-line connectors — rural-to-junction feeders

Output Artifact: All cluster assignments and features were saved to **data/trains_with_clusters.csv** (3,572 rows x 28 columns).

8. INTERPRETATION OF RESULTS & INSIGHTS

8.1 Key Insight #1 — Data Quality Beats Algorithm Complexity

Fixing the duration_m bug improved every single supervised model (up to +0.0739 F1) and the K-Means silhouette score (+0.0269). This is the strongest empirical finding of the project: understanding and repairing data quality yields greater gains than switching algorithms.

8.2 Key Insight #2 — Decision Tree Excels on Rule-Bound Data

The Decision Tree achieved 91.50% accuracy, 0.8896 Macro F1, and 0.8743 +/- 0.0155 CV F1 — outperforming KNN, SVM, and Naive Bayes. Indian Railways classifies trains via hierarchical threshold rules, which axis-aligned tree splits model naturally. Top features: third_ac (35.5%), distance (22.3%), total_duration (18.0%).

8.3 Key Insight #3 — Complementary Unsupervised Taxonomy

K-Means (k=5) uncovered interpretable regional/premium archetypes, while DBSCAN isolated 0.4% operational outliers. Together, they provide a data-driven taxonomy of Indian Railways trains that complements administrative zone boundaries.

8.4 Why Naive Bayes Failed

Gaussian Naive Bayes performed poorly (37.02% accuracy) because its core assumption — feature independence — is heavily violated. Distance, Duration, and AC coach configurations are strongly correlated ($r > 0.85$).

9. LIMITATIONS

- **1. Moderate class imbalance:** The minority Superfast class (16.1%) is harder to classify than Passenger (55.1%).
- **2. 8.50% error rate:** Dominant confusion occurs between Express and Superfast due to boundary overlap.
- **3. No live telemetry:** Data is a static timetable snapshot rather than real-time NTES running data.
- **4. Feature scope:** Halt count, average commercial speed, delay history, and seasonality are not included.
- **5. Zone one-hot sparsity:** 19 binary zone columns dilute Euclidean distances for KNN and SVM.

- **6. No ensemble methods:** Random Forest, XGBoost, and Gradient Boosting were kept as future scope.

10. FUTURE ENHANCEMENTS

- **1. Ensemble & Gradient Boosting:** Random Forest / XGBoost / LightGBM to resolve the 8.50% Express <-> Superfast overlap.
- **2. Live NTES Telemetry:** Incorporate real-time delay data, halt count, and average commercial speed.
- **3. REST API Deployment:** Deploy persisted joblib models (dt_classifier.joblib & scaler.joblib) as a REST API.
- **4. Deep Learning Baseline:** Evaluate TabNet and Multi-Layer Perceptrons (MLP) with zone embeddings.
- **5. Web Dashboard:** Interactive dashboard for zone-wise train tier predictions.
- **6. Time-Series Forecasting:** Predict train volumes by zone and season.

11. BIBLIOGRAPHY

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- **3.** Aurelien Geron — Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media.
- **4.** Tom M. Mitchell — Machine Learning, McGraw Hill Education.
- **5.** Sripaad Srinivasan — Indian Railways Dataset, Kaggle (https://www.kaggle.com/datasets/sripaadsrinivasan/indian-railways-dataset).
- **6.** scikit-learn Documentation — https://scikit-learn.org/stable/
- **7.** pandas Documentation — https://pandas.pydata.org/docs/
- **8.** Matplotlib Documentation — https://matplotlib.org/stable/

12. APPENDIX

12.1 Project Folder Structure

```
Indian-Railway-Train/
|-- data/
|   |-- trains.json                    # Original dataset (14.08 MB)
|   |-- trains.csv                     # Converted CSV (636 KB)
|   +-- trains_with_clusters.csv       # Cluster output (3,572 x 28)
|-- notebooks/
|   +-- main_notebook.ipynb            # Main notebook (all 4 phases)
|-- extras/
|   |-- advanced_ml.ipynb              # Model persistence, CV, error analysis
|   +-- README.md
|-- models/
|   |-- scaler.joblib
|   +-- dt_classifier.joblib
|-- visualizations/                    # Saved evaluation & EDA plots
|-- reports/
|   +-- Indian_Railway_ML_Presentation.pptx
|-- .gitignore
|-- README.md
+-- requirements.txt
```

12.2 Hardware & Software Requirements

Table 12.1: Hardware and Software Specifications

Component	Specification
OS	Windows 10 / Windows 11
Python	3.13.6
Editor	Jupyter Notebook
Libraries	pandas, numpy, scikit-learn, matplotlib, seaborn, joblib
RAM	Minimum 4 GB
Storage	500 MB free space

12.3 Reproducibility & GitHub Repository

```
pip install -r requirements.txt
# Open notebooks/main_notebook.ipynb in Jupyter Notebook and run all cells top-to-bottom

Repository URL : https://github.com/rajputpri/Indian-Railway-Train
Branch         : master
```

12.4 Justification for Skipped Topics

- **Linear Regression — Not Applicable:** The target variable type is categorical (3 classes: Pass, Exp, SF), not continuous. Linear Regression predicts numeric values and would produce meaningless outputs (e.g., Pass = 0.87). The correct approach for categorical targets is classification, which is why we used KNN, Decision Tree, Naive Bayes, and SVM.
- **Apriori — Not Applicable:** Apriori discovers association rules from transactional/basket data (e.g., customers who buy X also buy Y). This dataset contains operational features per train, not transactions. Therefore, K-Means and DBSCAN were chosen for unsupervised pattern discovery.